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Sex differences in latent cognitive abilities ages 6 to 59: Evidence from the Woodcock–Johnson III tests of cognitive abilities ☆

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Abstract

Sex differences in the latent general and broad cognitive abilities underlying the Woodcock–Johnson Tests of Cognitive Abilities were investigated for children, youth, and adults ages 6 through 59. A developmental, multiple indicator–multiple cause, structural equation model was used to investigate sex differences in latent cognitive abilities as well as developmental changes in these differences across the 6 to 59 age span. Females showed a consistent advantage on the latent processing speed (Gs) factor, and males showed a small, consistent advantage on the latent comprehension–knowledge (Gc) factor. Males also showed an advantage on latent quantitative reasoning (RQ) and visual–spatial ability (Gv) factors at most ages, although the latter was statistically significant only for adults. No statistically significant sex differences were shown on latent auditory processing, short-term memory, long-term retrieval, or fluid reasoning factors. The higher-order, latent *g* factor showed inconsistent differences for children, small, non-significant differences favoring females for adolescents, and fairly consistent statistically significant differences favoring females in adulthood. Findings are inconsistent with developmental theory that suggests males should show an advantage on *g* in adulthood. Supplemental analyses suggested that methodological choices, including the use of latent variables versus composites and methods for dealing with missing data, can affect research findings.
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1. Introduction

Despite considerable attention focused to the topic of sex differences in intelligence, a number of inconsistencies concerning the presence, direction, and magnitude of the differences remain. The identification of the ages at which differences typically emerge as well as the pattern of differences from childhood through adulthood is fundamental for understanding the true nature of sex differences in cognitive abilities. The current study examined differences in latent cognitive abilities between males and females on the Woodcock–Johnson III Tests of Cognitive Abilities, while taking into account the developmental changes across ages 6 to 59.

1.1. Sex differences in broad cognitive abilities

Males and females have shown consistent differences in some broad cognitive abilities across the lifespan. In

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general, females have shown an advantage in processing speed (Gs) from childhood through adulthood (Burns & Nettelbeck, 2005; Camarata & Woodcock, 2006; Hedges & Nowell, 1995). Males, in contrast, generally have demonstrated an advantage in broad visual–spatial ability (Gv) and related subtests (Jensen, 1998; Lynn, Fergusson, & Horwood, 2005; Maitland, Intrieri, Schaie, & Willis, 2000; Reynolds, Keith, Ridley, & Patel, in press; Rosén, 1995). Recent findings examining sex differences in some aspects of visual–spatial ability have not favored either sex, however (Camarata & Woodcock, 2006). Thus, it appears that there may be sex differences in specific aspects of spatial ability, such as spatial perception, mental rotation, spatial visualization, or visual memory (Linn & Peterson, 1985; Voyer, Voyer, & Bryden, 1995).

The evidence for sex differences in fluid reasoning (Gf) is mixed. Some evidence suggests that there are no practical differences between males and females from childhood through adulthood on aspects of Gf (Camarata & Woodcock, 2006; Kaufman & Horn, 1996). If fluid reasoning is measured by Progressive Matrices Tests, however, findings from recent studies suggest that adult males show an advantage over adult females on these measures (Lynn, Allik, & Irwing, 2004; Lynn & Irwing, 2004). Males have also shown an advantage on quantitative reasoning (RQ) (Benbow & Stanley, 1980; Jensen, 1998), a component of fluid reasoning (Carroll, 1993; Keith, 2005).

Studies examining crystallized ability (also known as comprehension–knowledge) at the broad ability, composite scale, and subtest level have been inconsistent, with findings depending on the nature of the task. For instance, females have shown an advantage in general verbal ability subtests and word knowledge, whereas males have shown an advantage in verbal analogies and general knowledge (Hyde & Linn, 1988; Lynn, Irwing, & Cammock, 2001). Recent research examining a latent crystallized ability (Gc) factor suggested that when general intelligence was controlled, boys showed a statistically significant advantage through the age of 16 (Reynolds et al., in press).

Less is known about the presence of sex differences in other broad abilities, including short-term memory, long-term memory, and auditory processing; findings from existing studies are also inconsistent. For example, Camarata and Woodcock (2006) reported no sex differences in short-term memory (Gsm), long-term retrieval (Glr), and auditory processing (Ga) composites from childhood through adulthood. A comparison of latent Gsm and Glr factors also showed no significant sex differences when controlling for *g* (Reynolds et al., in press). However, results from other studies suggest differences favoring females in short-term memory (Jensen, 1998; Maitland et al., 2000), long-term memory (Hedges & Nowell, 1995; Johnson & Bouchard, 2007a), and auditory processing (Hulick, 1998).

1.2. Sex differences in general intelligence

Some researchers have argued that there is no significant sex difference in general intelligence (e.g., Jensen, 1998; Mackintosh, 1996), and results from several recent studies have supported this position (e.g., Camarata & Woodcock, 2006; Deary, Strand, Smith, & Fernandes, 2007; Jensen, 1998; van der Sluis et al., 2007; van der Sluis et al., 2006). In contrast, Lynn (1999) has argued that for adults, males should demonstrate an advantage in general intelligence, and several recent findings have supported this position (Deary, Irwing, Der, & Bates, 2006; Irwing & Lynn, 2005; Jackson & Rushton, 2006; Lynn & Irwing, 2004; Lynn, Raine, Venables, Mednick, & Irwing, 2005; Nyborg, 2005).

According to Lynn, sex differences in intelligence need to be considered within the context of human development. Lynn suggested that results from previous studies of sex differences that have been based on children or young adults mask true differences in general intelligence due to differences in maturation (1999). There is a positive correlation between brain size and sex (McDaniel, 2005; Vernon, Wickett, Bazana, & Stelmack, 2000), and males have larger brains, on average, than do women. Thus, if this correlation is the result of a causal process, and one that accounts for between group differences as well as within group differences, then adult males should have an advantage over females in general intelligence. Lynn hypothesized that the male advantage in general intelligence should not emerge until late adolescence or early adulthood because females mature at an earlier age than do males. Specifically, Lynn's developmental theory of intelligence proposed that there should be small or negligible sex differences in general intelligence from ages 8 to 15, but by age 16 males should begin to show an increasing advantage through adulthood. Lynn's theory (1994, 1999) is noteworthy because it is an attempt to resolve an anomaly in the literature: males have larger brains than females, yet there is no observed difference in intelligence between males and females (Ankney, 1992; Rushton, 1992). Lynn has since demonstrated empirical support for his theory, although the ages at which differences emerged varied to some degree (Colom & Lynn, 2004; Lynn, Allik, & Must, 2000). Nevertheless, questions remain regarding how Lynn's developmental

theory fits into understanding cognitive ability differences in males and females.

Lynn's proposed developmental theory of intelligence is especially important in that he suggested that developmental differences between boys and girls may obfuscate true differences in adult general intelligence. Although some findings from studies of psychometric IQ have supported this theory, other studies applying latent variable methodological techniques have not (Dolan et al., 2006; Härnqvist, 1997; Reynolds et al., in press; van der Sluis et al., 2007; van der Sluis et al., 2006). We believe that latent variable models of intelligence provide a more detailed and accurate picture of sex differences, and may provide more insight into the nature and development of male and female differences in intellectual ability. Here, we investigate possible changes in sex differences in cognitive ability that are due to differences in age.

1.3. Methodological considerations

There are several possible reasons for the inconsistencies throughout the literature on sex differences in cognitive abilities. As already noted, sex differences may emerge or diminish at different ages (Lynn, 1999). It is thus important to control for age in analyses where developmental trends are likely. As suggested by Lynn (1999), cross-sectional studies of intelligence need to use data disaggregated by age so that each ability is assessed at each age level.

Another possible reason for inconsistent findings is the use of *emergent* versus *latent* variable definitions of constructs. Many researchers have used subtests or composite scores as estimates of the constructs being studied, whereas others have defined constructs via factor scores or components from factor analysis or principal components analysis (see Nyborg, 2003 for a review of studies). Fewer studies have examined differences in latent variables (Dolan et al., 2006; Härnqvist, 1997; Reynolds et al., in press; Rosén, 1995; van der Sluis et al., 2007; van der Sluis et al., 2006). Intellectual abilities, including *g*, Gv, and others, are generally conceived as latent variables, that is, traits of the individual that are only imperfectly measured via various subtests designed to measure those abilities. Composite scores (e.g., a Full Scale IQ) are necessary to approximate one's score on these latent variables at the individual level, and they may give hints as to sex differences at the group level. Composites, also known as emergent variables, are confounded with unreliability and invalidity, however. As a result, subtest invalidities are added together, for example, in the creation of a visual–spatial (Gv) composite, and unique variance (i.e., invalidity) due to broad abilities is added together into a Full Scale/IQ composite. As a result, apparent sex differences in such composites may be due to true differences in the constructs, or such differences may be due to invalidity. This confounding occurs whether traditional composites are used (e.g., a Full Scale IQ) or whether composites are created based on factor analysis (e.g., a factor score based on an unrotated first factor). In contrast, analyses using latent variables remove the effects of error and unique variance from estimates of the relations among constructs, and thus more accurately estimate sex differences in the true constructs of interest. Differences in findings may thus be due, in part, to the use of emergent rather than latent variables.

Tests used in analyses of sex differences should sample the broad intellectual abilities being studied adequately and validly. There is evidence, for example, that the Arithmetic test of the Wechsler series measures quantitative reasoning (RQ) (Keith, Fine, Reynolds, Taub, & Kranzler, 2006) or quantitative knowledge (Gq) (Phelps, McGrew, Knopik, & Ford, 2005) as much or more so than short-term/working memory (Gsm). Thus, analyses that use Arithmetic only in a memory factor or composite have likely confounded sex differences due to quantitative skills with those due to working memory. Relatedly, research and tests based on well-validated theories of intelligence are more likely to produce accurate outcomes than those based on an atheoretical approach.

1.4. Purpose and research questions

The purpose of this research was to determine the presence and magnitude of mean differences in latent broad and general cognitive abilities across the sexes. This research also sought to determine whether there are developmental changes in these differences from

Table 1
Sample sizes for the WJ III sample, by demographic characteristics

Variables	Sex	
	Male	Female
Race		
White	2688	2890
African American	407	533
Native American	65	59
Asian/Pacific Islander	168	156
Hispanic descent	400	391
Non-Hispanic descent	2930	3249
Total	3330	3640

Table 2
Description of the Woodcock–Johnson III tests used in analyses

Test name	Broad ability	Narrow ability	Task requirements
Concept formation	Gf	Induction	Determine the rule for the difference between the geometric shapes that are inside a box from those outside the box
Analysis synthesis	Gf	General sequential reasoning	Analyze and solve a series of increasingly complex puzzles using a given colored key to identify the missing components
Number series	RQ (Gf)	Quantitative reasoning	Complete a numerical sequence by applying the underlying rule or principle
Number matrices	RQ (Gf)	Quantitative reasoning	Determine the relations among numbers within a numerical pattern to complete an analogy
Spatial relations	Gv	Visualization, spatial relations	Identify the pieces needed to form a complete shape from a series of shapes
Picture recognition	Gv	Visual memory	Recognize a subset of previously displayed pictures within a distracting, larger set of pictures
Block rotation	Gv	Visualization, spatial relations	Identify two of the presented three-dimensional visual patterns and match the one that has been rotated in spatial orientation
Visual matching	Gs	Perceptual speed	Rapidly find and circle two identical numbers from a set of six numbers with the difficulty of the items increasing
Decision speed	Gs	Speed of processing, semantic processing speed	Quickly scan a row of seven pictures and then identify the two that have the most in common
Cross out	Gs	Perceptual speed	Quickly find and mark identical pictures from a larger set of pictures
Visual auditory learning	Glr	Associative memory meaningful memory	Learn and recall graphic pictures representing words (rebuses)
Memory for names	Glr	Associative memory	Recall names that were previously learned
Sound blending	Ga	Phonetic coding	Listen to a word presented in syllables or phonemes and synthesize the sounds together to produce a whole word
Incomplete words	Ga	Phonetic coding	Listen to a word that is missing phonemes and identify the complete word
Sound Patterns–voice	Ga	Sound discrimination	Listen to a pair of voice-like sound patterns and determine the similarity or difference within the pair
Auditory attention	Ga	Speech–sound discrimination resistance to auditory stimulus distortion	Indicate which picture corresponds to the word presented from an audio recording with increasingly distracting background noise
Numbers reversed	Gsm	Working memory	Hold a series of numbers in immediate awareness and repeat the sequence in reverse order
Auditory working memory	Gsm	Working memory	Retain a mixed series of numbers and words in immediate awareness while reorganizing into two sequences of words, and then numbers, in the same order presented
Memory for words	Gsm	Memory span	Repeat a list of orally presented words that are unrelated in content in the correct sequence
Verbal comprehension	Gc	Lexical knowledge language development	Identify pictured objects, provide antonyms and synonyms for given words, and complete verbal analogies
General information	Gc	General (verbal) information	Determine where objects are usually found and what people usually do with an object
Story recall	Gc	Listening ability, meaningful memory	Recall elements of stories presented orally
Academic knowledge	Gc	General information, science/ geography/cultural information	Answer questions related to science, social studies, and humanities
Picture vocabulary	Gc	Language development lexical knowledge	Provide names for pictured objects from more common to less well-known items
Oral comprehension	Gc	Listening ability	Identify the missing keyword that makes sense after listening to a short passage

childhood through adulthood. To do so, we used data from a nationally representative U.S. standardization sample from a well-validated, theory-derived measure of intelligence. We explicitly tested for developmental changes in sex differences across a 6 to 59 year age span. Structural equation modeling was used to examine the magnitude of sex differences in latent, rather than emergent, variables, using a well-validated hierarchical

Table 3
Summary statistics for the 25 WJ III cognitive tests, by sex

Subtest	Male					Female				
	N	*M*	SD	Skew	Kurtosis	*N*	*M*	SD	Skew	Kurtosis
Academic knowledge	2927	103.944	14.486	−.084	.242	3229	101.413	13.556	−.091	1.697
Analysis synthesis	2281	102.450	15.370	−.062	.668	2498	101.990	13.688	−.114	1.014
Auditory attention	2472	102.761	12.552	−.152	3.330	2656	103.537	11.725	−.080	1.710
Auditory working memory	1799	101.917	15.201	−.185	.999	2014	103.376	13.458	−.225	.534
Block rotation	1915	102.475	17.600	−.031	3.120	2166	99.117	15.722	.333	3.335
Concept formation	3031	100.750	14.966	−.150	.025	3362	102.136	14.009	−.239	.539
Cross out	2534	100.332	14.239	.035	.604	2861	103.027	14.030	.025	.517
Decision speed	2748	98.931	15.096	.092	.553	3011	103.723	15.122	.028	1.134
General information	2660	102.286	14.953	−.339	.596	2873	101.325	13.883	−.373	.944
Incomplete words	2590	102.121	14.928	−.216	2.649	2928	102.129	14.690	−.175	1.962
Memory for names	2156	103.239	15.607	.414	.769	2451	100.603	14.963	.409	1.064
Memory for words	2990	101.232	15.270	.054	.367	3281	101.557	14.629	.020	.605
Number matrices	2648	103.041	14.779	−.026	−.094	2966	101.038	14.518	−.003	.497
Number series	3003	103.401	16.184	−.347	1.751	3204	101.368	14.075	−.248	1.350
Numbers reversed	2699	100.881	15.664	−.052	.643	2902	101.579	14.802	−.003	.519
Oral comprehension	2998	101.482	15.327	−.040	.631	3232	102.342	14.481	−.099	1.529
Picture recognition	2219	98.994	13.596	−.012	.492	2428	103.439	14.633	.010	.631
Picture vocabulary	3005	103.929	15.534	−.046	.513	3303	101.188	14.582	.091	.752
Sound blending	2960	101.101	15.462	.199	.274	3245	102.250	14.712	.086	.118
Sound patterns	2398	103.146	16.334	.037	.227	2651	99.371	15.670	.252	1.271
Spatial relations	2603	101.459	14.829	.076	.425	2790	100.476	14.499	−.139	1.109
Story recall	2875	104.561	13.023	−.107	.266	3270	104.094	13.077	−.008	.822
Verbal comprehension	2234	103.278	15.108	−.237	.085	2433	102.254	14.367	−.177	.716
Visual auditory learning	2179	103.748	13.874	.396	.702	2470	103.917	13.854	.299	.275
Visual matching	2932	98.821	14.877	−.104	1.430	3246	104.006	14.510	.032	1.225
g[a]	3330	99.411	15.620	−.309	.306	3640	100.539	14.390	−.378	.925

[a]*g* estimate is based on an model-derived imputed *g* score, based on the model shown in Fig. 4. The mean and standard deviation were set to 100 and 15, respectively, for the total sample.

theory of intelligence that is consistent with the measure used. The research focused on three primary questions:

1. Are there statistically significant and meaningful sex differences in mean levels of the latent broad cognitive abilities measured in this research?
2. Is there a statistically significant and meaningful sex difference in mean levels of a latent higher-order general (*g*) factor?
3. Do any such sex differences vary developmentally?

2. Method

2.1. Participants

Participants were drawn from the standardization sample of the Woodcock–Johnson III (WJ III) (Woodcock, McGrew, & Mather, 2001). The WJ III was standardized on a nationally representative U.S. sample of 8818 children and adults, ages 2 through 90; the sample was stratified, based on census projections for 2000, on race, sex, educational level, occupational status, community size, and SES. For this research, data were used from all WJ III standardization sample participants who were between 6 and 59 years of age. We chose this age range because it spans from school-age through adulthood, including ages at which developmental changes in sex differences are likely. Younger children and older adults were not included to minimize potentially confounding developmental questions concerning the structure of intelligence for younger children and the possibility of health-related declines in cognitive abilities for older adults (Carroll, 1993, chap 17). In addition, the youngest children in the WJ III standardization sample did not take all of the tests used in this research. Primary analyses used the total sample. Follow-up analyses used the total sample divided into 10 age levels: 6 to 7, 8 to 9, 10 to 11, 12 to 13, 14 to 15, 16 to 17, 18 to 22, 23 to 29, 30 to 44, and 45 to 59. Table 1 shows sample sizes and descriptive information for these participants.

2.2. Instrument

The full Woodcock–Johnson III includes 53 cognitive and achievement tests across three batteries: the Tests of Cognitive Abilities, the Tests of Achievement, and the

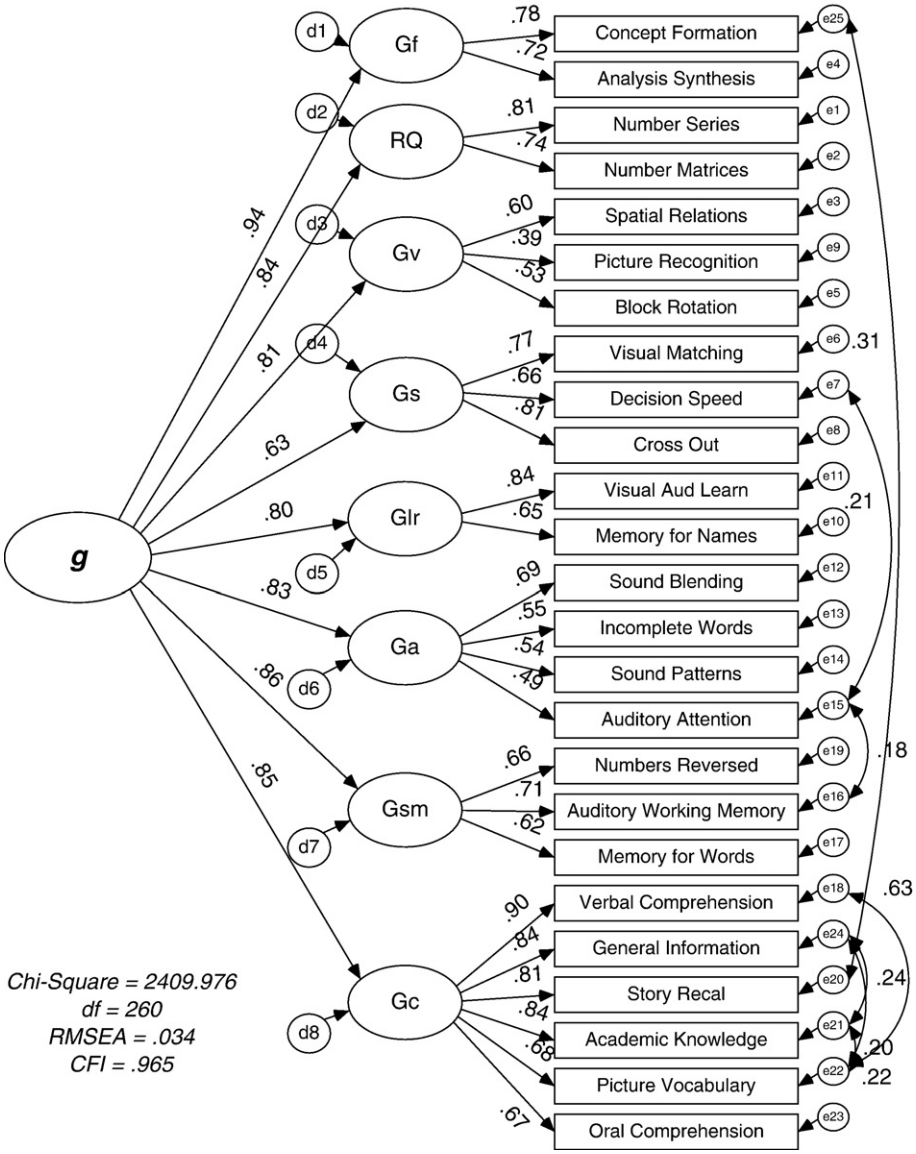

Fig. 1. Initial higher-order model for the WJ III Cog, across ages, with one higher-order factor and eight first-order factors.

Diagnostic Supplement to the Tests of Cognitive Abilities. Most of the tests used in this research were from the WJ III Cognitive Battery.

The WJ III Cognitive battery was designed specifically to measure at least seven of the broad cognitive abilities from Cattell–Horn–Carroll (CHC) theory, a combination of Cattell and Horn's Gf–Gc theory and Carroll's three-stratum theory (cf. McGrew, 2005; McGrew & Woodcock, 2001): comprehension–knowledge (Gc), long-term retrieval (Glr), visual–spatial ability (Gv), auditory processing (Ga), fluid reasoning (Gf), processing speed (Gs), and short-term memory (Gsm). The WJ III is designed to measure at least two narrow abilities for each broad ability, as well as Carroll's third stratum, *g*. Because previous research has suggested a sex difference in quantitative reasoning (symbolized by Carroll as RQ, 1993, p. 197), RQ was modeled here as a separate factor, rather than as a narrow ability subsumed under Gf. Research has supported the structure of the WJ III across ages (McGrew & Woodcock, 2001; Taub & McGrew, 2004), and independent reviewers have been impressed with the test's development, standardization, and psychometric properties (Cizek, 2003; Sandoval, 2003). The 25 tests used in this research are briefly

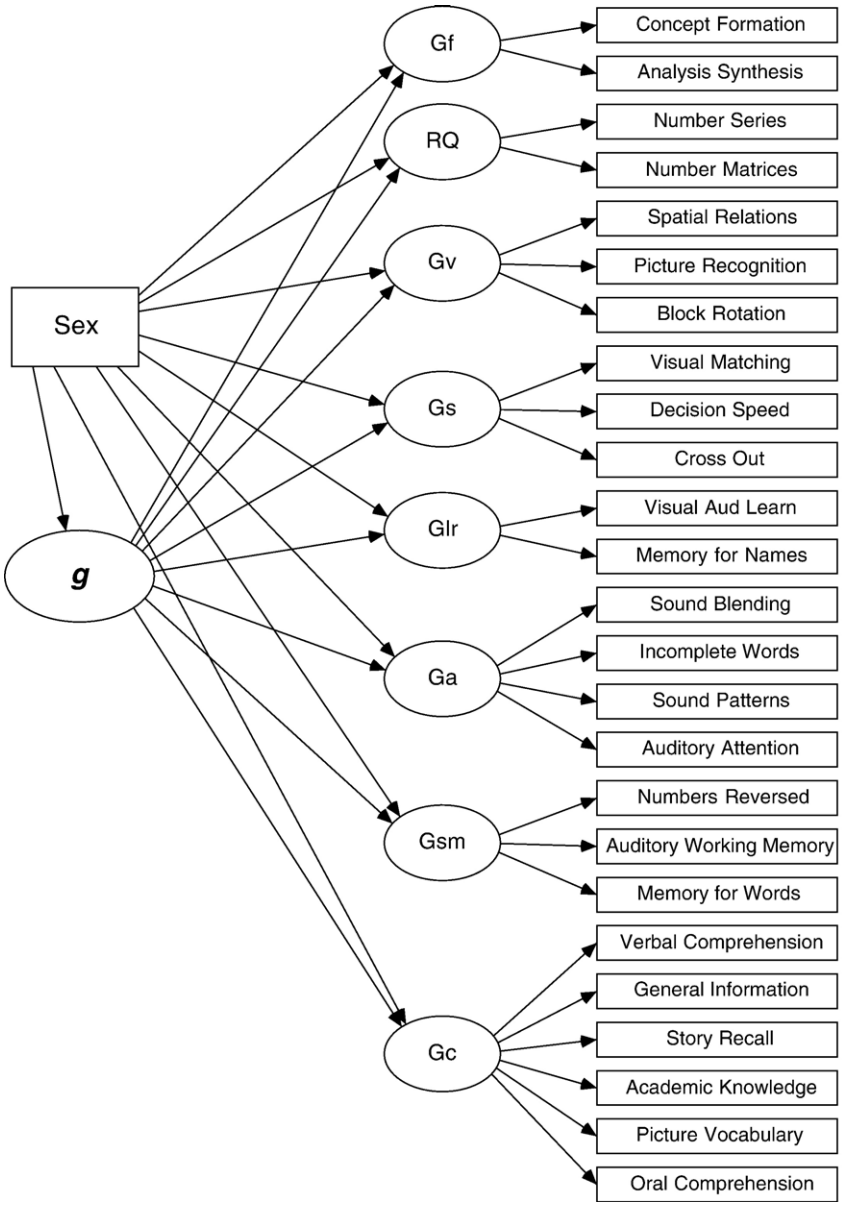

Fig. 2. Example of a MIMIC model for estimating sex differences in latent first-and higher-order abilities.

described in Table 2; for more information concerning the broad and narrow abilities measured by each test, see Schrank (2005).

2.3. Analytic approach

SPSS was used for initial data preparation and descriptive analyses. The Amos 5 (Arbuckle, 2003) and Amos 7 (Arbuckle, 2006) programs were used to conduct all structural equations analyses. The analyses will be briefly discussed here, and presented in more detail in the results.

As the first step in the analyses, we sought to develop a higher-order confirmatory factor model that was consistent with CHC theory and with previous research, and that also fit the data reasonably well. Possible sex differences in the latent general and broad abilities were tested using multiple indicator–multiple cause (MIMIC) models, one of two possible SEM approaches for testing differences in latent means. Prior to MIMIC analysis, we tested whether the data met the assumptions necessary for this method and also allowed for partial intercept invariance by allowing, when needed, effects directly

Table 4
Tests of the equality of variance/covariance matrices for males and females, by age level

Age level	RMSEA	CFI	*n*
Total	.017	.995	6970
6 to 7	.034	.974	643
8 to 9	.024	.986	964
10 to 11	.024	.989	1007
12 to 13	.033	.981	676
14 to 15	.021	.994	594
16 to 17	.025	.990	557
18 to 22	.024	.989	1011
23 to 29	.040	.976	491
30 to 44	.042	.977	600
45 to 59	.054	.963	427

from sex to subtests. The primary analyses used a series of MIMIC models that also included an interaction term that tested for differential sex differences across the 6 to 59 year age span. Post-hoc probing of statistically significant sex by age interactions used multi-group models across 10 age levels to allow for variation in sex differences across ages.

The root mean square error of approximation (RMSEA) (Steiger, 1990) and the comparative fit index (CFI) (Bentler, 1990) were used to asses the fit of single models. For analyses comparing the fit of competing, nested models, $\Delta\chi^2$ was used as the primary measure of fit, supplemented by the Akaike Information Criterion (AIC) (Akaike, 1987). Individual parameters were also compared to their standard errors as a test of statistical significance. For all substantive questions, a probability of .01 was used.

3. Results

Preliminary analyses examined descriptive statistics for the age groups and tests used. The sample sizes, means, standard deviations, skew, and kurtosis for each test by sex are shown in Table 3. At this stage we also developed a higher-order CFA model across all age levels. The goal of these preliminary analyses was to measure adequately each of the seven broad CHC abilities, along with the narrow ability of RQ. Thus, we began with factor structures examined in previous research (McGrew & Woodcock, 2001; Taub & McGrew, 2004), but with a slightly different orientation. Rather than focusing on the *best* factor structure for all WJ III tests (McGrew & Woodcock, 2001) or testing the theoretical factor structure for a limited number of tests (Taub & McGrew, 2004), our primary interest was to achieve strong representation of the broad abilities so that we could examine differences in those broad abilities for males and females. To this end, we focused on developing a reasonable fitting model with a minimum of cross-loadings. This strategy resulted in the dropping of several tests that likely measured more than one ability. We also freed large correlated errors, as suggested by modification indices[1].

The resulting preliminary model is shown in Fig. 1. Using conventional criteria (Hu & Bentler, 1998, 1999), the model fit well across the entire 6 to 59 year age span. Most factors appear well-measured, although Glr was only represented by two tasks, both of which measure, in part, the same narrow ability (associative memory). Two other potential Glr tasks were dropped from the model because they showed cross-loadings with other factors. In contrast, the Gf and RQ factors appear adequately represented by only two tests apiece.

Higher-order multiple indicator–multiple cause (MIMIC) models were the primary method used to test for mean sex differences in latent abilities. For this model, sex was a dummy-coded exogenous variable with potential paths to each of the broad abilities and to *g*, and with unstandardized path coefficients representing the mean difference across the sexes. The alternative for such analyses is multiple group analysis of mean and covariance structures (MG-MACS). MG-MACS models are more flexible because, unlike MIMIC models, MG-MACS models do not assume that males and females share a common covariance structure (Kano, 2001). If the variance/covariance matrices for males and females are equivalent, however (and thus factor loadings, residual variances, factor variances, and covariances are equivalent across groups), the two approaches produce the same estimates of mean differences across the groups (Muthén, 1989; Reynolds et al., in press). The MIMIC approach, however, tests fewer parameters (and is thus more parsimonious), is easier to interpret, and allows for the inclusion of continuous covariates, such as age. An example of a MIMIC model is shown in Fig. 2.

For the total sample and for each age group, we initially tested the equivalence of the variance/covariance matrices across the sexes in a multi-group model. As shown in Table 4, for the total sample and for each age group, the model specifying the equality of the variance–covariance matrices fit the data well (Hu & Bentler, 1999). These analyses thus suggested that the variance/covariance matrices are equivalent for males

[1] For preliminary analyses using modification indices, we analyzed covariance matrices produced using full information maximum likelihood (FIML) methods. For all other analyses, including all SEMs shown here, we used raw data, consisting of age-based standard scores. For all analyses, FIML methods were used to deal with missing data.

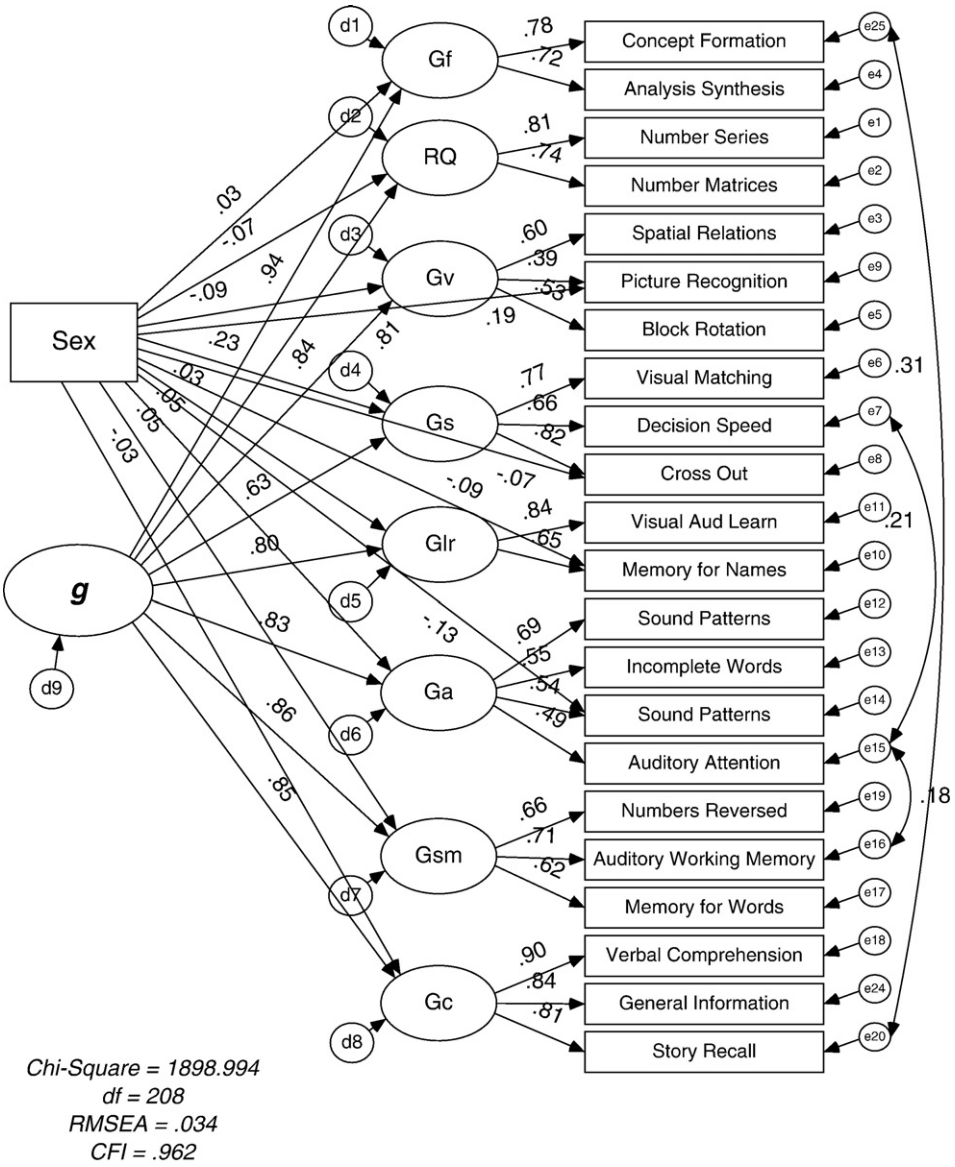

Fig. 3. Partial intercept invariance across ages. Paths drawn from Sex to tests allow sex differences on tests above and beyond differences on *g* and broad abilities.

and females, and that a MIMIC model could be used to examine sex differences. Previous research has supported the factorial invariance of the WJ III across age levels, although that research used fewer tests than were included in our models (Taub & McGrew, 2004) or previous versions of the WJ (Bickley, Keith, & Wolfle, 1995). A model specifying the equality of the variance–covariance matrices across the 10 age groups used here fit quite well according to one criterion (RMSEA = .043) and adequately according to the other (CFI = .937). This analysis suggested that the MIMIC model could also be used across the age levels.

A single MIMIC model across all ages was used to test for intercept (or scalar) invariance (i.e., the equivalence across the sexes on the measured variables, given possible differences in the first-order factors). For this analysis, the Sex to *g* path was set to zero, whereas the means of the first-order factors were allowed to vary (i.e., the Sex to broad ability paths were estimated). Modification indices were used to free any necessary paths from Sex to measured variables. We freed such paths sequentially if they had modification indices of 10.8 or larger, corresponding to a probability of approximately .001. *Partial* scalar invariance was

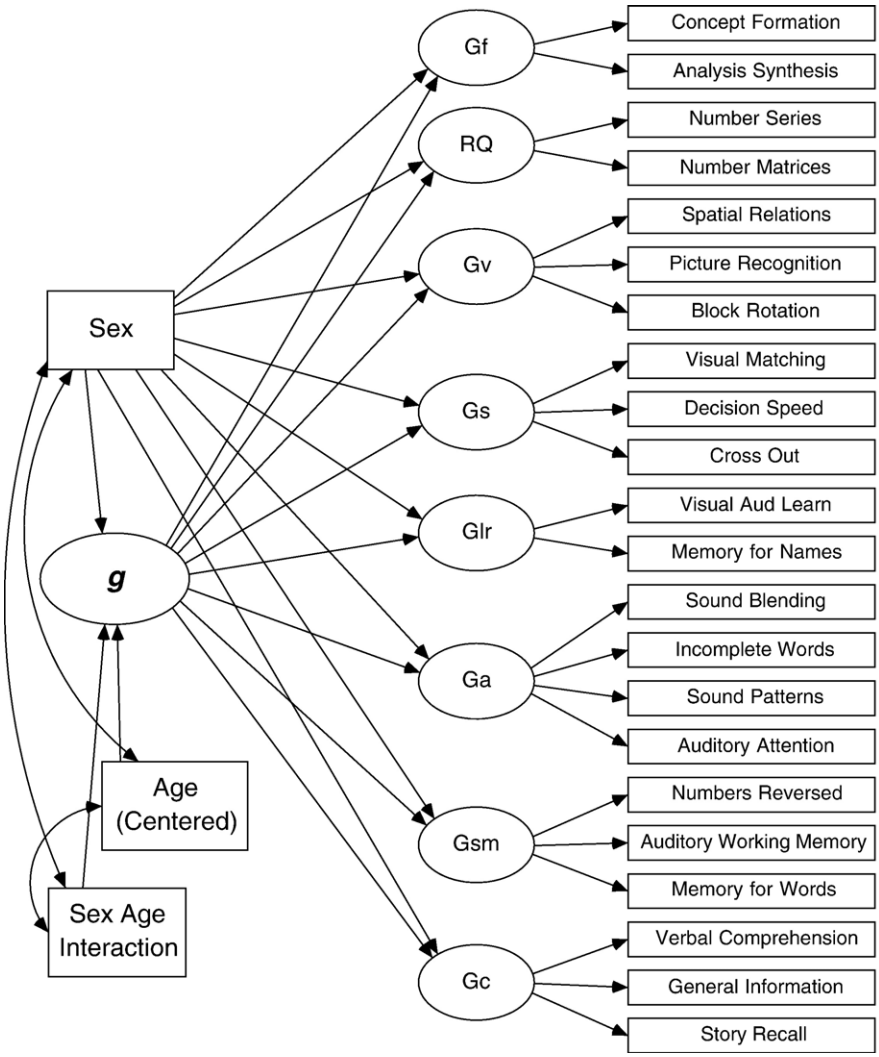

Fig. 4. Testing for developmental differences in latent cognitive abilities. Statistically significant effects from Sex * Age Interaction to *g* and first-order factors represent developmental changes in mean differences across ages.

established with the addition of seven such paths, meaning that seven subtest intercepts were allowed to vary across the sexes. Given the large number of parameters tested and the large sample size (and resulting power), this level of partial intercept invariance is not unusual (Byrne, Shavelson, & Muthén, 1989). Nevertheless, because three of these differences in intercepts were for tests on the Gc factor (Academic Knowledge, Picture Vocabulary, Oral Comprehension), and because the factor was well-measured without these three tests, the three tests were excluded for the remainder of analyses for this model. Thus the final partial invariance model as shown in Fig. 3 used 22 tests and allowed for four subtest intercept differences across the sexes. Partial invariance of this sort suggests that these four subtests showed differences across sex above and beyond those on the latent ability represented by the factor. Such differences likely reflect sex differences on narrow abilities not represented at the factor level.[2]

3.1. Primary analyses

With the establishment of the equality of variance–covariance matrices and partial scalar invariance, the substantive questions of this research were evaluated. An

[2] We also tested all models with the original 25 tests with virtually identical results as those reported here. In addition, we tested our final models without allowing these differences in intercepts, with near identical results.

Table 5
Comparison of primary models testing sex differences across ages

Model	χ^2 (*df*)	$\Delta\chi^2$ (*df*)	*p*	AIC	RMSEA	CFI
1. Sex differences for *g*	3129.668 (257)			3315.668	.040	.944
2. *g*+Ga differences	3109.843 (254)	19.826 (3)	<.001	3301.843	.040	.944
3. *g*+Gc differences	2864.742 (254)	264.927(3)	<.001	3056.742	.038	.949
4. *g*+Gf differences	3095.995 (254)	33.673 (3)	<.001	3287.995	.040	.945
5. *g*+Glr differences	3120.249 (254)	9.419 (3)	.024	3312.249	.040	.944
6. *g*+Gs differences	2822.981 (254)	306.688 (3)	<.001	3014.981	.038	.950
7. *g*+Gsm differences	3118.812 (254)	10.856 (3)	.013	3310.812	.040	.944
8. *g*+Gv differences	3075.832 (254)	53.836 (3)	<.001	3267.832	.040	.945
9. *g*+RQ differences	2981.035 (254)	148.634 (3)	<.001	3173.035	.039	.947
10. *g*+other differences	2415.185 (246)	714.483 (11)	<.001	2623.185	.036	.958
11. Fix NS age differences	2426.536 (249)	11.351 (3)	.010	2628.537	.035	.958
12. Fix NS sex differences	2427.638 (250)	1.103 (1)	.294	2627.638	.035	.958
13. Fix *g* interaction	2442.666 (251)	15.028 (1)	<.001	2640.666	.035	.957
14. Fix *g* age difference	2427.920 (251)	.281 (1)	.596	2625.920	.035	.958
15. Fix *g* mean difference	2426.381 (252)	8.461 (1)	.004	2632.381	.035	.957

Note: Models 2 through 10 are compared to model 1. Models 11 through 13 are compared to the previous model. Model 14 is compared to model 12, and model 15 is compared to model 14. Model 14 is the final model interpreted in the text.

initial, simplified, model is shown in Fig. 4. Note that the model includes additional influences beyond sex. Because one purpose of this research was to analyze potential developmental changes in sex differences, the model also includes an influence from Age (centered, in years) and a Sex by Age interaction term.[3] Thus in this model, an effect for Sex on *g* represented a sex difference in general intelligence. Because the Sex variable was coded 0 for males and 1 for females, a positive path would represent an advantage for females, whereas a negative path would represent an advantage for males. In this model, a statistically significant path from the Sex * Age interaction term to an ability would mean that there was statistically significant variation in sex differences in that broad ability across the ages. The depiction of the model shown in the figure is simplified in that the potential paths from Age and from the Sex * Age interaction term to the general and broad abilities are not shown (errors and paths allowing differences in subtest intercepts are also not shown).

The initial model included effects for the three exogenous/difference variables (Sex, Age, and Sex * Age) on *g* (model 1 in Table 5). The fit statistics associated with the various models are shown in Table 5. Subsequent models added the effects of the exogenous variables on each of the broad abilities. These models tested whether *g* accounts for *all* sex differences across groups, or whether there are also sex-related differences in, for example, RQ. Each broad ability was examined, in turn (models 2 through 9), and those paths that were statistically significant ($p<.01$, as judged by the *z* values, that is, path/standard error of the path) were combined into a single model (model 10). Such a strategy was necessary because a model with all paths drawn to both *g* and all broad abilities would be under-identified. Several of the paths that were statistically significant when examined in isolation were not significant when examined in combination (model 10); these were deleted in groups. Our analysis plan called for the initial deletion of non-significant ($p>.01$) Sex * Age to ability paths, but all interaction paths that were statistically significant in models 2 through 9 were also statistically significant when examined in combination. Three Age effects that were no longer statistically significant were deleted (i.e., fixed to zero, model 11), as was one broad ability mean difference (model 12). Again, the fit statistics associated with these models are shown in Table 5, although as noted, the statistical significance of individual parameters was the primary criterion by which paths were retained or deleted.

Table 6
Sex differences and sex by age interactions across the 6 through 59 year age span (primary analysis)

Ability	M_{diff}	Sex * Age interaction
g	1.21	$p<.01$
Gf		
RQ	−3.21	$p<.001$
Gv	−3.58	$p<.01$
Gs	6.01	
Glr		
Ga		
Gsm		
Gc	−2.07	

[3] The model only included linear Age effects. Follow-up analyses suggested no need to include nonlinear Age effects (e.g., Age squared).

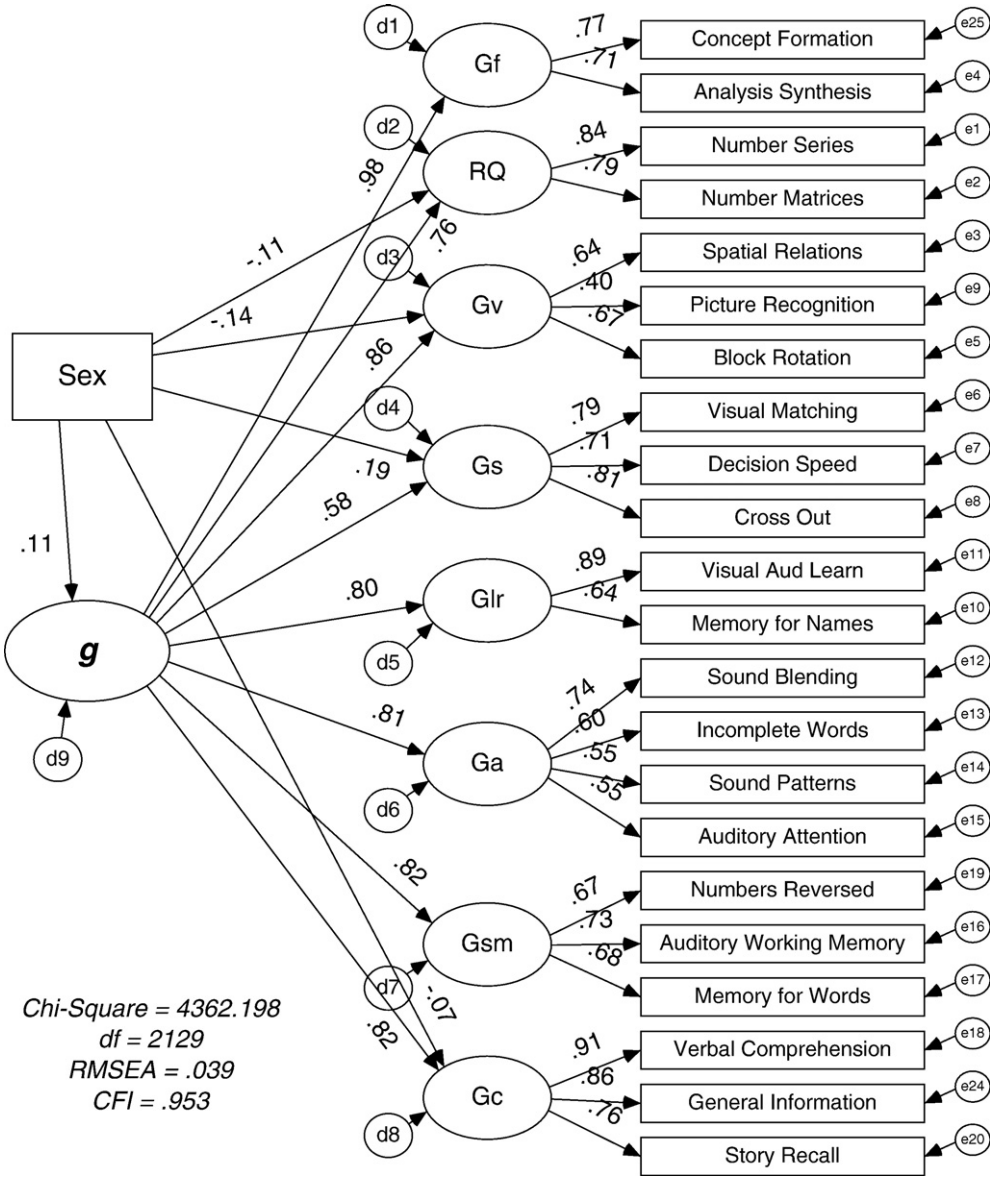

Fig. 5. Post-hoc probing of developmental changes across 10 age groups. Results shown are for ages 18 to 22; the fit statistics pertain to the entire model across all ages.

In models 1 through 12 in Table 5, the paths from the three exogenous variables to *g* were estimated, thus allowing sex differences in *g* and variations in those differences across ages. In model 12, the path from the interaction term to *g* was statistically significant ($p<.01$) as was the path from Sex to *g*, whereas the path from Age to *g* was not significant. In other words, there were statistically significant mean differences in *g* with sex differences in the unique aspects of the broad abilities controlled, but there was also significant variation in those differences across the ages. These differences were also tested in models 13 through 15. As shown in the Table, constraining the Sex * Age interaction for *g* to zero resulted in a statistically significant loss of fit ($\Delta\chi^2=15.028$, $p<.001$), as did constraining the sex difference on *g* to zero ($\Delta\chi^2=8.461$, $p=.004$). Model 14 is considered the final model, and included paths from the Sex variable to (i.e., sex differences on) the latent *g*, RQ, Gv, Gs, and Gc factors. This model also included statistically significant Sex by Age interactions for *g*, RQ, and Gv. This finding means that the extent of the sex differences varied across ages for these latent variables.

Table 7
Sex differences in *g* and broad abilities, by age, for the higher-order model

Age group	Gs	Gc	RQ	Gv	*g*
6 to 7	6.49*	−2.24*	1.12	−1.76	−1.94
8 to 9	5.88*	−2.10*	−3.73*	−2.88	.49
10 to 11	5.74*	−1.94*	−1.79	−1.77	−1.33
12 to 13	5.67*	−1.89*	−3.94*	−2.05	.86
14 to 15	5.73*	−1.70*	−2.19	−.74	.78
16 to 17	6.07*	−2.09*	−3.92*	−2.93	.95
18 to 22	5.63*	−2.11*	−3.48*	−4.24*	3.46*
23 to 29	6.04*	−1.98*	−3.91*	−4.02*	1.08
30 to 44	6.58*	−2.00*	−1.66	−7.18*	3.52*
45 to 59	4.91*	−1.92*	−7.86*	−9.39*	4.80*

*$p<.01$.

Table 6 summarizes the extent of these differences from the primary analyses. The second column in the table shows the sex differences in general and broad cognitive abilities that were statistically significant, and the magnitude of those differences in standard IQ points.[4] Positive differences favor females and negative differences favor males. The third column shows those Sex * Age interactions that were statistically significant ($p<.01$, based on the z values).

As shown in the table, females showed a statistically significant advantage on the latent processing speed (Gs) factor of 6 IQ points, with differences in the latent *g* factor taken into account. Females also showed a small but statistically significant advantage on the latent *g* factor, with unique differences due to the broad abilities taken into account. This difference was equivalent to 1.21 IQ points, but the statistically significant interaction shows that there were also developmental changes in this difference across the 6 to 59 age span. In contrast, males showed a consistent 2 point IQ advantage on a latent comprehension–knowledge (Gc) factor after controlling for differences in *g*. Males also showed a statistically significant advantage for quantitative reasoning (RQ) and visual–spatial ability (Gv), although the statistically significant Sex * Age interactions for these broad abilities show that these differences varied across age.

[4] These were calculated using the standardized paths and the variance of the Sex variable. For the total sample, the *SD* of Sex was .50. Thus, the difference between males (coded 0) and females (1) was 2.00 *SD* units. The standardized coefficient from Sex to Gs was .20, and thus the difference between males and females on the processing speed broad ability (controlling for general intelligence) was .40 standard deviation units. Conversion to a standard IQ scale (*SD*= 15) results in a difference between males and females of 6.00 (more exactly, 6.009) IQ points on Gs, favoring females.

3.2. Probing statistically significant sex by age interactions

Statistically significant Sex * Age interactions were evidence for developmental changes in sex differences. We probed those interactions using multi-group MIMIC analysis across 10 age groups (the age groups and sample sizes are the same as shown in Table 4). For this analysis, a MIMIC model was estimated across each of 10 age groups. In this model, paths were estimated from Sex to any ability that showed a statistically significant sex difference or a statistically significant Sex * Age interaction. These analyses allowed the estimation of sex differences by age for abilities that the primary analysis had shown to have developmental changes in sex differences (viz., *g*, RQ, and Gv). For those abilities that showed a statistically significant main effect but a non-significant interaction, the unstandardized Sex to ability paths were constrained to be equal across all age groups. All other parameters were allowed to vary across age groups. The results of this analysis for ages 18 to 22 are shown in Fig. 5, which also shows the fit of this model across all age levels. Again, the model showed a good fit to the data using conventional criteria.

Table 7 shows the primary findings from the post-hoc probing: mean differences (in IQ points) for all significant differences, by age group. These findings are also displayed visually in Fig. 6. Again, the sex differences were converted to a common IQ metric for this graph. As shown in Table 7 and Fig. 6, and noted earlier, females showed a consistent advantage of around 6 IQ points on a latent processing speed (Gs) variable, after sex differences in latent *g* were taken into account. Said differently, these are the effects of Sex on Processing Speed above and beyond the effects of Sex on *g*. As noted, because the primary analysis showed no developmental changes for this sex difference, the effect was constrained equal across age groups; there are minor differences in effects because it is the unstandardized coefficients that were constrained to equality and the figures take into account the possible differences in the variances of the latent factors (i.e., they are based on standardized coefficients). All ages are starred in the table because the overall analysis showed this effect to be statistically significant with no significant variation across ages.

Males showed a consistent advantage of approximately 2 IQ points on the latent comprehension–knowledge (Gc) factor, as shown in Table 7 and Fig. 6. Like Gs, Gc showed no statistically significant interaction, and thus this parameter was constrained to be equal across the age groups.

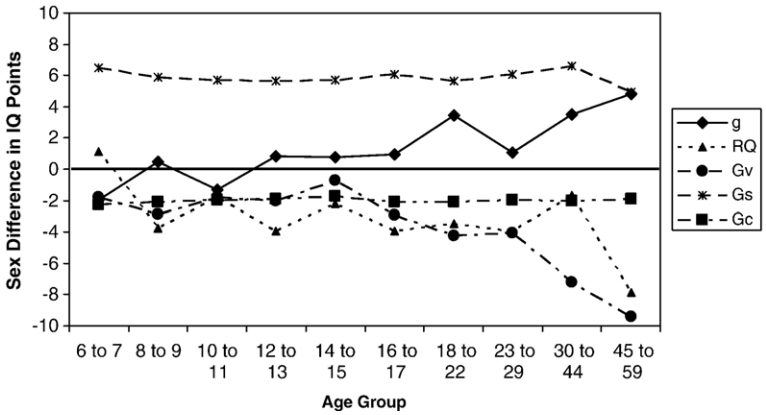

Fig. 6. Sex differences on the latent broad and general factors across 10 age groups. Positive differences favor females, whereas negative differences favor males. Differences are in standard IQ points.

The results for the quantitative reasoning (RQ) latent variable are also shown in Fig. 6 and Table 7. Because the primary analysis showed a statistically significant Sex * Age interaction for the male/female mean difference on RQ, the path from Sex to RQ was allowed to vary across the age groups in this post-hoc probing. As shown in the Table and Figure, there was resulting variability in the sex differences in quantitative reasoning across ages. Nevertheless, consistent with previous research, RQ showed a fairly consistent advantage favoring males; that male advantage was statistically significant ($p<.01$) at 6 out of 10 age groups. The sex difference ranged from a slight female advantage at ages 6–7 to a 7.9 point male advantage for the oldest age, but was generally about 3 to 4 points favoring males.

Males scored at a higher level than females on the latent visual–spatial ability (Gv) factor (controlling for *g*), but this difference only became statistically significant at the 18–22 age level. After age 18, the male advantage ranged from approximately 4 to 9 IQ points; this difference and trend is shown in Fig. 6 (and Table 7).

Table 7 and Fig. 6 also show the findings for the latent *g* factor. As shown in the Table, at younger age levels, there were small, inconsistent, non-significant differences in latent *g*. These differences stabilized and began favoring females at about age 12, but did not become statistically significant until ages 18–22. Even after age 18, not all effects were statistically significant, although they all favored females. These findings thus suggest either no difference in a latent *g* factor, or a small difference (approximately 2 IQ points) favoring females, with that difference appearing to increase with age. It is important to note that these were the differences on a latent *g* factor with the unique sex differences associated with the other broad abilities controlled. Said differently, these are the effects of Sex on *g* controlling for the significant effects of Sex on four of the broad abilities.

3.3. Nested factors model

Our research has focused on a higher-order model of intelligence consistent with the instrument used and the theory from which it was derived. Others have argued for the superiority of an alternative hierarchical model, the nested factors or bifactor model (Gustafsson & Balke, 1993). An example of such a model with these data is shown in Fig. 7. The nested factors model has several potential advantages compared to the higher-order model. First, it is agnostic about the relation of *G* to the broad abilities, meaning that fewer assumptions are made about the structure of intelligence. Instead, the nested factors model places *G* and the broad abilities in the same factor space (this is also why the general factor is symbolized as *G* rather than *g* in this model). Because the nested factors model is agnostic about the relation of *G* to the broad abilities, it is conceptually similar to partialing out the effects of an unrotated first factor prior to group factor analysis (e.g., Johnson & Bouchard, 2007a). Because *G* and the broad abilities occupy the same factor space, it is possible to estimate the effects of Sex on both the broad abilities and on *G* simultaneously, rather than using the incremental approach used in our analyses with the higher-order model. Although the nested factors and higher-order models appear quite different, it is in fact possible to go from one to another via an intermediary model that places proportionality

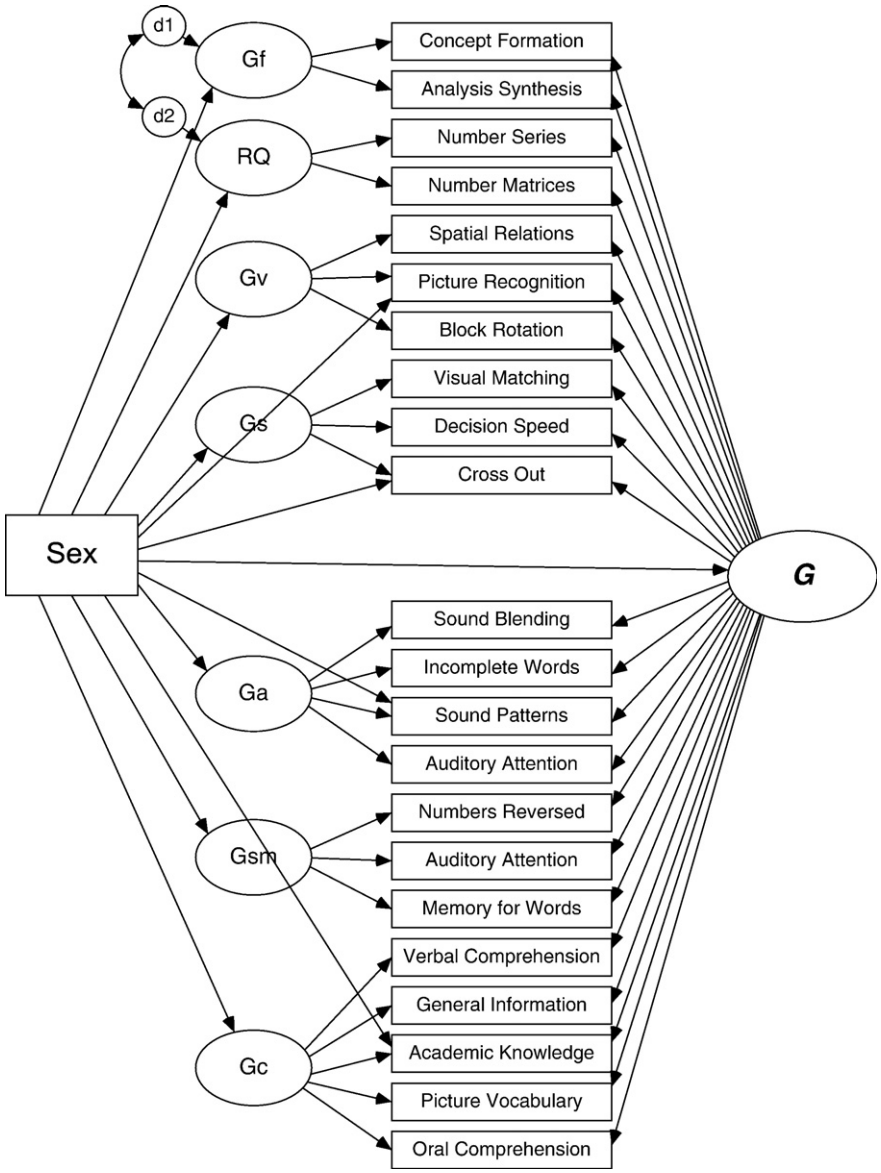

Fig. 7. A nested factors model for estimating sex effects in latent cognitive abilities.

constraints on the loadings of the tests on the broad abilities (Yung, Thissen, & McLeod, 1999). That is, the *g* in the higher-order model, by affecting the tests only through the broad abilities, affects each subtest proportionally based on the loading of the broad abilities on the higher-order *g* factor.

There are also several disadvantages to the nested models approach. It is often difficult to estimate the nested factors model because of overparameterization, and additional constraints, not theoretically derived, are sometimes required to make the model work. Because the broad abilities are generally left uncorrelated with this approach, factors usually require three or more subtests or they will be empirically under-identified. Finally, and most importantly from our perspective, we believe that the higher-order model is a better representation of cognitive abilities and of most hierarchical theories of intelligence, with *g*, the most general ability, affecting less general cognitive abilities which, in turn, affect more narrow cognitive abilities. Despite our preference for the higher-order model, we believed it important to estimate a nested factors version of our model to examine possible consistencies and inconsistencies in findings.

A nested factors version of the MIMIC model tested in this research is shown in Fig. 7. We went through a

Table 8
Sex differences and Sex * Age interactions for the nested factors model

Ability	M_{diff}	Sex * Age interaction
g	3.51	$p<.001$
Gf	−6.22	
RQ	−8.65	$p<.001$
Gv	−8.26	$p<.001$
Gs	5.83	
Ga	−2.34	
Gsm	−3.75	
Gc	−6.97	

similar developmental process for this model as we did the higher-order model, first estimating an initial model using all subtests along with the Sex variable, then freeing needed intercepts (three were freed), and then testing the model across ages using a Sex, Age, and Sex * Age interaction term. As noted earlier, adjustments are needed to avoid empirical underidentification in such models. We deleted the Glr factor and the subtests that loaded on that factor. We also deleted the Story Recall subtest because it was a general source of misfit in the model. We achieved identification of the Gf and RQ factors by allowing their unique variances to be correlated, suggesting that they share something in common beyond the other first-order factors in the model. We believe this model relaxation to be valid because RQ is most likely a narrow ability subsumed under a broader Gf factor (Carroll, 1993; Keith, 2005). The errors of measurement are not shown in the figure to simplify the display; there were also two correlated errors.

The model fit the data well (RMSEA = .035, CFI = .967). Table 8 shows the results of this overall analysis, with the Sex difference main effects converted to standard IQ points. As shown in the table, all of the broad abilities, and *G*, had statistically significant main effects for Sex, meaning that they showed significant sex differences ($p<.01$). What is most striking about the table is that all such ability differences were in the same direction as in the previous, higher-order model estimation. Such effects were also generally larger, and sex differences were also found for fluid reasoning (Gf, favoring males), short-term memory (Gsm, favoring males), and auditory processing (Ga, favoring males).

As noted in the table, several of these ability mean differences were accompanied by Sex * Age interactions, meaning that the sex differences shown in the main effects varied across the age groups. Post-hoc probing of this model across the 10 age groups required additional model constraints for model identification (the subtest error variances were constrained to be equal across the groups, and Gf factor loadings were constrained to be equal across ages 16–29), but the final post-hoc multi-group model fit the data reasonably well despite these constraints (RMSEA = .051, CFI = .930). The results from these analyses are shown in Table 9. Again, all differences were in the same direction as in the higher-order model, and generally larger. The nested factors approach would lead to the same conclusions about Gs, Gv, Gc, RQ, and *G* as the higher-order model. In the post-hoc probing none of the Ga or Gsm sex differences were statistically significant.

4. Discussion and conclusions

Previous research has suggested the presence of sex differences on several intellectual abilities (processing speed, quantitative reasoning, visual–spatial ability), but findings have been inconsistent for other broad cognitive abilities as well as for general intelligence. Here we used structural equation modeling to test for sex differences in general intelligence and eight other broad cognitive abilities and to examine developmental changes in those sex differences across a 6 to 59 year age span. Primary

Table 9
Sex differences in *G* and broad abilities, by age (nested factors model)

Age group	*g*	Gf	RQ	Gv	Gs	Ga	Gsm	Gc
6 to 7	−.30	−3.17*	.00	−3.08	6.81*	−4.34	−2.24	−6.41*
8 to 9	1.69	−2.25*	−6.34*	−3.94	5.65*	−1.17	−1.85	−5.34*
10 to 11	−.19	−3.30*	−3.44*	−3.30	6.08*	−1.45	−1.13	−5.32*
12 to 13	1.79	−4.41*	−6.14*	−4.57	5.66*	−1.13	−1.38	−5.40*
14 to 15	1.97	−4.08*	−5.37*	−3.11	5.87*	−1.31	−.94	−6.23*
16 to 17	2.79	−5.67*	−9.71*	−4.22	6.18*	−1.10	−1.18	−5.80*
18 to 22	5.09*	−13.47*	−6.79*	−10.26*	5.93*	−1.75	−1.11	−5.05*
23 to 29	2.00	−6.24*	−6.02*	−10.83	6.61*	−2.12	−.97	−4.69*
30 to 44	4.57*	−4.68*	−3.93	−15.03*	8.21*	−2.73	−1.90	−7.22*
45 to 59	6.28*	−5.23*	−13.99*	−17.46*	7.31*	−1.84	−0.54	−6.70*

*$p<.01$.

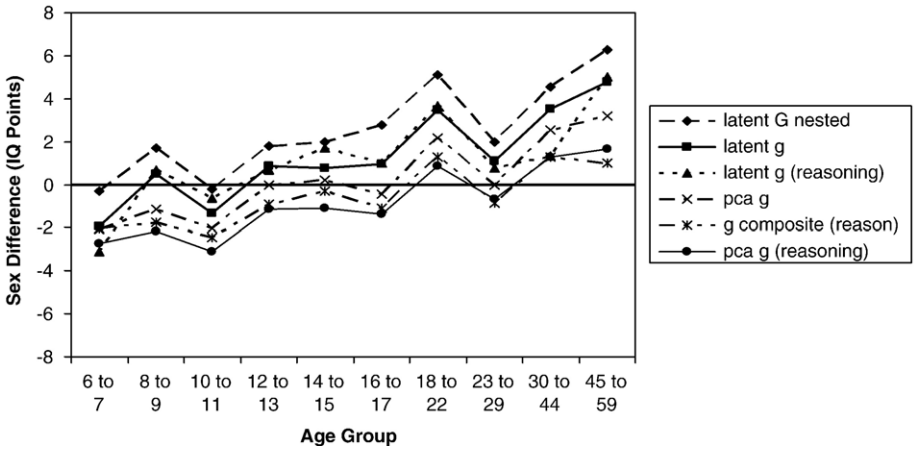

Fig. 8. Sex differences in psychometric *g* as a result of different methods of estimating *g*. Methods treating *g* as an *emergent*, rather than *latent*, variable were more dependent on the measures used to create *g* and were more likely to favor males.

analyses showed consistent sex differences in latent means across the ages for two broad abilities, processing speed (favoring females) and comprehension–knowledge (favoring males). The findings also showed significant sex differences in latent means for quantitative reasoning and visual spatial ability (both favoring males) and psychometric *g* (favoring females). The latter three differences varied across the age levels, however. Post-hoc probing of these interactions showed that the sex difference in quantitative reasoning was fairly consistent across the ages, again favoring males. The difference in visual spatial ability favoring males was also seen fairly consistently across the ages, but appeared to increase with age and only became statistically significant at age 18.

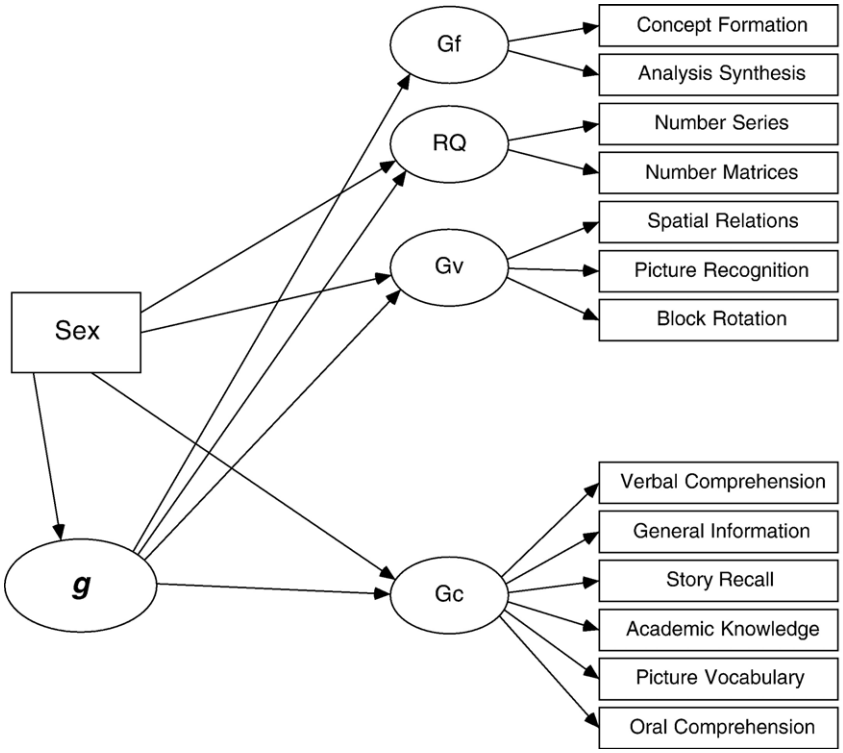

Fig. 9. A latent MIMIC model based only on reasoning factors and tests. Results for sex differences on this latent *g* were similar to those from the primary analyses.

There were no significant differences in *g* at younger ages, but at later ages, small differences were found, favoring females by 2 to 4 IQ points.

4.1. Caveats and supplemental analyses

Perhaps the most surprising finding of this research is the small mean difference favoring adult females on the latent *g* factor. This finding is inconsistent with Lynn's developmental theory suggesting that, beginning at approximately age 16, males should show about a 4 point advantage in global IQ over females (Lynn, 1999). The present research showed a statistically significant Age * Sex interaction for *g*. Post-hoc probing of that interaction suggested little or no difference in *g* across the sexes for children and adolescents, but small sex differences favoring females starting at age 18. There are several possibilities for these incongruous findings, including different definitions of constructs, different components of *g*, different tests, and different samples. We discuss some of these possibilities, and empirically test some of them, below.

One possibility, of course, is that these findings are simply a fluke. Perhaps the tests that make up the WJ III are different enough from other measures of intelligence, or the WJ III sample skewed enough, so that the test shows a difference in one direction when, in fact, real differences are in the other direction. Several lines of evidence argue against this possibility. First, all validity evidence suggests that the WJ III is an excellent measure of CHC abilities and *g* (Cizek, 2003; Sandoval, 2003; Woodcock et al., 2001) and that the *g* measured by the WJ III is the same *g* measured by other instruments (Keith, Kranzler, & Flanagan, 2001). By all standards, the standardization of the WJ III was also excellent. Furthermore, these findings are also consistent with one other recent study using similar latent variable methods, but using a different sample and a different test (Reynolds et al., in press). Finally, logically, if a skewed sample or skewed tests were responsible for our findings for *g*, our findings for the broad abilities would also likely be different than expected. Nevertheless, additional research using these methods with other samples and tests is needed.

4.1.1. Emergent versus latent g?

Another possible reason for such differences is the use of a latent variable versus emergent variables or composites. As noted in the Introduction, we believe this is a crucial distinction in research on intelligence in general, and research on sex differences in intelligence, in particular. This research used SEM methods to examine sex differences in latent general and broad cognitive abilities, with variance due to error and that due to narrow, specific abilities removed from both the broad abilities and from *g*. In addition, and perhaps critically, this methodology also removes the unique broad ability variance (e.g., variation due to a broad Gv factor) from consideration of sex differences in *g*. The examination of sex differences in a latent *g* gets closer to a true general factor, with that general factor less affected by the tests that reference the broad abilities and by the broad abilities that reference *g*. In contrast, much of the other research on sex differences has used composite scores to estimate *g* and various broad abilities. Such emergent variables (composites) add together error and specific variances of subtests in the creation of broad ability composites, and variance unique to the broad abilities is added together into a *g* composite. Thus, emergent composites add together differences due, for

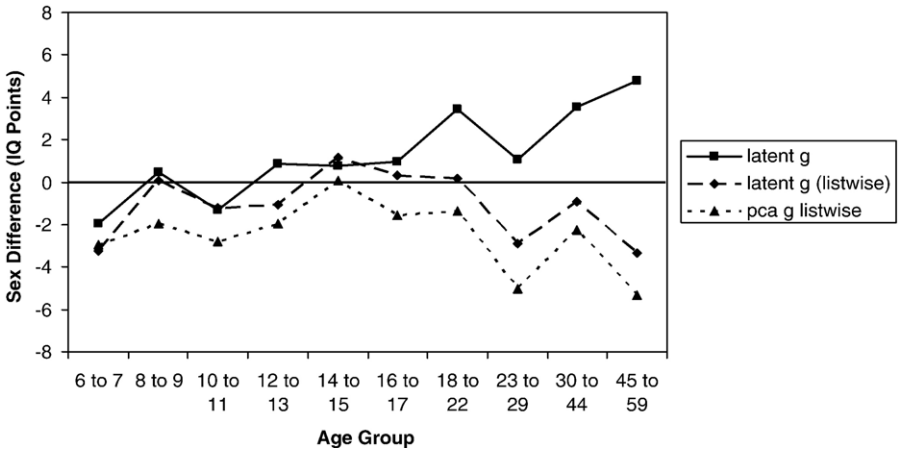

Fig. 10. Sex differences in psychometric *g* as a result of different methods of dealing with missing data.

example, to Gv, RQ, and other broad abilities into a *g* composite, whereas latent variables examine *g* cleansed of this unique variance. Either approach is defensible if it is consistent with theory. Most intelligence theories, however, conceptualize general intelligence and broad abilities as latent, not emergent, variables. It is important to note that any composite variable, any *g* variable created from adding subtests or scales together in various combinations, is still an emergent variable, even if the weighting of that composite was dictated by principal components analysis, factor analysis, or structural equation modeling. A composite is a composite, and a composite is not a latent variable.

We conducted several analyses to determine if our findings related to *g* would differ if we used a *g* composite rather than a latent *g* variable. As will be shown, the use of composites versus latent variables would affect our findings. First, we computed a *g* composite as an unrotated first component from a principal components analysis of the 25 WJ III tests used in this research. Fig. 8 shows mean differences in *g* across the ages with *g* estimated using different methods. The top two lines show the average sex differences in *G* from the nested factors model and *g* in the primary, higher-order model, respectively. The line labeled "pca *g*" in the graph shows sex differences on a *g* composite estimated using principal components analysis. When *g* is conceived as such a composite rather than a latent variable, males outscored females on *g* at several younger ages, and the female advantage was slightly smaller at the older ages.

In a related concern, critics might argue that our results are a function of the broad abilities used as indicators of *g*. Here, our research used a model based on the three-stratum/CHC theory of intelligence. This theory and model includes more broad abilities (e.g., Ga, Gsm) than do the models used in most other studies. For example, Lynn argued that general intelligence should be "defined as the sum of the verbal comprehension, reasoning and spatial abilities" (Lynn, 1999, p. 2). Thus Lynn's definition focused on a composite variable rather than a latent variable, and also on a more narrow definition of general intelligence. Perhaps our findings of inconsistent differences favoring females on *g* are a result of a less-reasoning-based, broader definition of *g*. This concern is related to the previous concern (latent versus emergent variables) because while sex differences in emergent composites will indeed differ, perhaps markedly, depending on the makeup of the composites, latent variables should be much more stable across definitions of the components of intelligence. That is, a latent *g* should be fairly invariant across definitions of intelligence, whereas a composite *g* will change depending on the variables used in the composite. To examine this concern, we conducted a MIMIC analysis across age groups using only "reasoning" tests to define broad abilities and *g*. The model is shown in Fig. 9 (correlated errors and partial intercept invariance paths are not shown). The results of this analysis were consistent with the other higher-order latent ability analyses. Indeed, as shown by the line labeled "Latent *g* (reasoning)" in Fig. 8, the sex differences by age for this analysis were very similar (with the exception of age 30–44) to those shown for the primary analyses (the "latent *g*" line). In contrast, when the tests shown in Fig. 9 were averaged to create Gf, RQ, Gv, and Gc *composites*, and those composites were further averaged to create a *g*-like composite, the sex difference line was noticeably lower, meaning it was more likely than previous models to favor males [the line is labeled "*g* composite (reason)" in Fig. 8]. Very similar results were shown when a composite was created using the first unrotated component of these reasoning tests [the line labeled "pca *g* (reasoning)" in Fig. 8].

These findings suggest one plausible explanation for differences in our findings concerning sex differences in *g* from those showing differences favoring adult males: our research modeled *g* as a latent variable, whereas much of the research showing differences favoring males estimated *g* via a composite. Composite variables, in turn, add together differences in other broad abilities, whereas latent variables partial out the effects of those broad abilities. As a result, latent variable findings are also much less dependent on the particular tests and scales used to estimate both broad and general abilities.

4.1.2. Missing values

Another possible methodological reason for differences in findings is the treatment by researchers of missing data. All analyses reported so far used full-information maximum-likelihood methods to deal with missing data, among the most sophisticated of such methods, and allowing for the use of the *complete* dataset (Schafer & Graham, 2002). Other missing value options include mean substitution, listwise deletion, or pairwise deletion of missing data. With up to 53 tests for the total WJ III battery, not all tests were administered to all participants. In the current research, for example, 2854 out of 6970 total participants had complete data for all 25 tests used. To determine the possible effect of methods for dealing with missing data, we conducted the latent variable MIMIC analyses using only participants with complete data for the 25 WJ III tests. The sex differences in *g* for this model are shown (the middle line) in Fig. 10.

As shown in the figure, the use of listwise deletion of missing data resulted in either no differences across the sexes or a difference favoring males (for older participants). We also recreated the unrotated first principal component of all 25 tests, although this time using listwise deletion to deal with missing data. The bottom line in Fig. 10 shows the sex differences in this emergent variable estimate of *g*. Using this method, males outscored females for all ages except one (ages 14–15).

This series of analyses shows that methodological decisions can indeed affect the results of analyses of sex differences in intelligence. When *g*, a latent variable from most theoretical perspectives, was modeled as a latent variable, results suggested inconsistent, non-significant sex differences in *g* for children and youth, but small differences favoring females beginning at age 18. In contrast, when *g* was modeled as an emergent variable, a composite, the sex differences were smaller and less likely to favor females. As already noted, this finding may be because a latent higher-order variable examines the effect of *g* with any sex differences in unique, lower-order abilities accounted for at the proper stratum. With composites, such differences in unique abilities are likely added together. Treatment of missing data also affected the results of these analyses. Maximum-likelihood methods require less stringent assumptions than do more common approaches (e.g., pairwise deletion, listwise deletion, mean substitution), allow researchers to use all available data, and generally produce less biased parameter estimates even when important assumptions are not fulfilled (Muthén, Kaplan, & Hollis, 1987). (Assumptions of maximum-likelihood estimation are that data are missing at random and that variables are normally distributed.) When full-information maximum-likelihood estimation was used, results suggested small, inconsistent sex differences for children and youth, but differences favoring females beginning at age 18. When listwise deletion was used (a common method), results suggested small differences, this time favoring adult males. Finally, we should note that the only condition under which our analyses supported Lynn's developmental theory (Lynn, 1999) was when we used both listwise deletion of missing data and modeled *g* as a composite, rather than a latent variable.

4.1.3. College students

One potential problem with this research involves a large subsample of college students (most in the 18–22 year age group). The WJ III included a special college norm group, and thus these students' standard scores are based on a different standardization than for non-college students. Of the 1011 participants in the 18–22 age group, 628 were from the college sample, and 242 of 491 participants in the 23–29 age group were a part of the college sample. To determine if these norming differences affected the findings for the 18–22 and 23–29 year age group, we re-conducted the post-hoc (multi-sample by age group) analyses using the *W* scores rather than the standard scores. *W* scores are Rasch-based transformations of raw scores, and can be considered as equal-interval raw scores (cf. Woodcock et al., 2001). *W* scores would thus not be affected by whether a participant was in the primary norm sample or a special sample. When *W* scores were analyzed, the same pattern of broad and general ability differences were shown as when standard scores were used (the same abilities were statistically significant, in the same direction) for both the 18–22 and the 23–29 age groups.

4.1.4. Partial intercept invariance revisited

It is also possible that the paths drawn from Sex to several subtests (freeing their intercepts for partial intercept invariance) somehow changed the findings of this research. To test this possibility, we re-analyzed the post-hoc model with the relevant intercepts for the Gs and Gv factors (those factors for which differences were found) constrained to be equal across the sexes. For this analysis, all differences that were statistically significant in the primary analysis were in the same direction for this follow-up analysis. There would have been little change in the findings if we had not allowed for partial intercept invariance (the same abilities were statistically significant, in the same direction).

4.1.5. MIMIC models

As already noted, one of the assumptions of the MIMIC approach used here is that the groups compared share a common covariance structure. We tested this assumption using an omnibus test of the equivalence of variance/covariance matrices across sexes and across ages. A more powerful approach would be to test factor loadings, factor variances, and other parameters for strict factorial invariance using a multi-group approach. Although more powerful, this approach also tests numerous parameters (over 600 for cross-age invariance), thus increasing the possibility of Type I errors. For this reason, we decided to use the omnibus tests of invariance rather than more specific tests of invariance. We should note, however, that the alternative approach could lead to differences, likely minor, in our findings.

4.1.6. CHC theory

All of the latent variable models tested in this research were based on a three-stratum/CHC theory of

intelligence. We believe this orientation is justified because the WJ III was developed based on CHC theory, and because there is a considerable body of research directly or indirectly supporting the factor structure of the current version of the test (McGrew & Woodcock, 2001; Taub & McGrew, 2004) as well as the previous version (Bickley et al., 1995). Nevertheless, there are many other possible factor structures that were not evaluated in this research; such models may have produced a different pattern of sex differences than those shown here. Given the consistency in findings across the few different models tested here, and the numerous models tested in similar, previous research (Reynolds et al., in press), we suspect any differences in findings would be minor, however.

4.1.7. Item selection and deletion

Finally, the test development process could have had an unknown effect on these findings. Like most modern tests, the WJ III used a panel of reviewers to identify items that might include content sensitive to "women, individuals with disabilities, and cultural or linguistic minorities" (McGrew & Woodcock, 2001, p. 52). Although it is unlikely that the sensitivity analyses discarded enough items to affect the latent variables in this research, the overall effect of such procedures is simply unknown. Of course, this caveat applies to any sex difference research, since all tests go through some process for choosing and eliminating items. In addition, statistical analyses were conducted to isolate WJ III items biased based on sex, race, and Hispanic descent [although no items showed differential item functioning based on sex (Camarata & Woodcock, 2006, p. 233)]. Again, the cumulative effect of such bias analyses is unknown, although we agree with Jensen, that "… eliminating group biases generally improves test validity, and vice versa" (Jensen, 1992, p. 276).

4.2. Developmental theory

If our findings are borne out by future research, then a final possibility is that the prediction from Lynn's developmental theory that adult males should show higher intelligence than adult females is incorrect. The reasoning behind that prediction is that there is a well-established positive correlation between brain size and intelligence (McDaniel, 2005; Vernon et al., 2000) and it is also well-established that males have greater overall brain volume (Giedd, Castellanos, Rajapakse, Vaituzis, & Rapoport, 1997; Gilmore et al., 2007; Reiss, Abrams, Singer, Ross, & Denckla, 1996). Thus males should show higher levels of intelligence than females, although developmental differences in childhood and adolescence should obfuscate the male advantage for those ages.

Here, males did not demonstrate higher *g*, but did show advantages in more specific abilities. Perhaps male and female differences in cognitive abilities may be described better in terms of more specific aspects of brain anatomy or function rather than overall brain size (cf. Haier, Jung, Yeo, Head, & Alkire, 2005; Johnson & Bouchard, 2007b; Johnson, Jung, Colom, & Haier, 2007). For example, when controlling for overall brain size, there are also sex differences in the volume of various subcortical structures, many of which favor females (e.g. amygdala and caudate head) (Giedd et al., 1997; Szabó, Lancaster, Xiong, Cook, & Fox, 2003). Moreover, males and females may use different parts of their brains when performing the same mental ability tasks (Gur et al., 1999; Haier et al., 2005). Specifically, adult males tend to show stronger correlations between *gray* matter regions and intelligence, whereas adult females show higher correlations with *white* matter regions and intelligence. Despite females' smaller cranial capacity compared to males, they may have a more organized system in place for the use of white matter to compensate for their smaller cranial capacity and volume (Gur et al., 1999; Schmithorst & Holland, 2006). One hypothesis is that males may rely more on specialized functioning of cortical structures whereas females rely more on efficient and organized neural functioning (Schmithorst & Holland, 2006). Evidence from functional connectivity research suggests that as boys get older their brains become more functionally specialized (Schmithorst & Holland, 2006). The findings reported here showed that, bar ages 23–29, the male advantage in Gv became considerably larger with age and the female advantage in *g* became larger with age. This finding may reflect a greater reliance on localized cognitive functioning or within hemispheric connectivity for males, whereas females may show a greater reliance on interhemispheric connectivity (or generalized cognitive functioning).

Lynn's (1999) contribution is important because developmental differences are likely to continue to be very important to understanding sex differences in cognitive abilities. Consideration should be given to developmental differences in structure and function, including whether overall volume is related to the development of more specialized abilities (Jensen, 1998). Moreover, it is likely important to consider the relation between function and structure as some of these sex differences may actually serve to minimize observable cognitive differences between males and females (De Vries, 2004).

This study did not, of course, study brain structure or function, thus no conclusions can be made about neuroanatomical or neurophysiological differences between males and females. The consistency of this study's findings with a similar previous study using latent variables (Reynolds et al., in press), however, should provide a more precise understanding of differences in the latent traits underlying the cognitive performance of males and females. For example, it has been hypothesized that *g* masks the underlying dimensions on which sex differences lie (Johnson & Bouchard, 2007a). This general hypothesis was indeed supported in this study; the results would have been considerably different if our models had not considered possible sex differences in *g* along with possible sex differences in other abilities. If such sex differences in *g* were not allowed, sex differences in short-term memory (Gsm) and auditory processing (Ga) would have appeared in the primary analysis, and the Sex * Age interactions for RQ and Gv would not have been statistically significant. In addition, all sex differences favoring males would have diminished, whereas the sex difference favoring females (Gs) would have increased.[5] Thus, it is important to take sex differences in *g* into account (similar to partialing them out) when considering possible sex differences in other abilities (and vice-versa). The most efficient way of doing so is to model sex differences within a latent variable system that allows researchers to model differences among the *latent* cognitive abilities at the appropriate level within one model. Such modeling also allows for tests that the constructs being compared are equivalent across groups.

4.3. Summary

The Woodcock–Johnson Tests of Cognitive Abilities were used to investigate sex differences in latent cognitive abilities. Data from the nationally representative U.S. standardization sample for children, youth, and adults ages 6 to 59 were analyzed. A developmental MIMIC model was used to examine sex differences in *g* and eight other broad cognitive abilities across ages, and to test for developmental differences across ages. The results showed a moderate, consistent mean difference favoring females on the latent processing speed (Gs) factor, and a small, consistent difference favoring males on the latent comprehension–knowledge (Gc) factor. At most ages, males showed an advantage on a latent quantitative reasoning (RQ) factor and on a latent visual–spatial factor (Gv), although the Gv difference became statistically significant only at older ages. The latent *g* factor showed inconsistent differences for children, small non-significant differences favoring girls during adolescence, and fairly consistent differences favoring females during adulthood. No sex differences were found on short-term memory (Gsm), long-term retrieval (Glr), fluid reasoning (Gf), or auditory processing (Ga) latent variables.

[5] This change in magnitude is because Sex also has indirect effects on RQ, Gv, Gc, and Gs through *g*. The *total* effects of Sex on RQ, Gv, and Gs are thus smaller, whereas the total effect of Sex on Gs is larger.

References

Akaike, H. A. (1987). Factor analysis and AIC. *Psychometrika*, *52*, 317−332.

Ankney, C. D. (1992). Sex differences in relative brain size: The mismeasure of women, too? *Intelligence*, *16*, 329−336.

Arbuckle, J. L. (2003). *Amos 5.0 update to the Amos user's guide.* Chicago: Smallwaters.

Arbuckle, J. L. (2006). *Amos 7.0 user's guide.* Chicago: SPSS.

Benbow, C. P., & Stanley, J. C. (1980). Sex differences in mathematical ability: Fact or artifact? *Science*, *210*, 1262−1264.

Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, *107*, 238−246.

Bickley, P. G., Keith, T. Z., & Wolfle, L. M. (1995). The three-stratum theory of cognitive abilities: Test of the structure of intelligence across the life span. *Intelligence*, *20*, 309−328.

Burns, N. R., & Nettelbeck, T. (2005). Inspection time and speed of processing: Sex differences on perceptual speed but not IT. *Personality and Individual Differences*, *39*, 439−446.

Byrne, B. M., Shavelson, R. J., & Muthén, B. O. (1989). Testing for equivalence of factor covariance and means structures: The issue of partial measurement invariance. *Psychological Bulletin*, *105*, 456−466.

Camarata, S., & Woodcock, R. W. (2006). Sex differences in processing speed: Developmental effects in males and females. *Intelligence*, *34*, 231−252.

Carroll, J. B. (1993). *Human cognitive abilities: A survey of factor-analytic studies.* New York: Cambridge University Press.

Cizek, G. J. (2003). Review of the Woodcock–Johnson III. In B. S. Plake, J. C. Impara, & R. A. Spies (Eds.), *The fifteenth mental measurements yearbook* (pp. 1020−1024). Lincoln, NE: Buros Institute of Mental Measurements.

Colom, R., & Lynn, R. (2004). Testing the developmental theory of sex differences in intelligence on 12–18 year olds. *Personality and Individual Differences*, *36*, 75−82.

De Vries, G. J. (2004). Minireview: Sex differences in adult and developing brains: Compensation, compensation, compensation. *Endocrinology*, *145*, 1063−1068.

Deary, I. J., Irwing, P., Der, G., & Bates, T. C. (2006). Brother–sister differences in the g factor in intelligence: Analysis of full, opposite-sex siblings from the NLSY1979. *Intelligence*. doi:10.1016/j.intell.2006.1009.1003

Deary, I. J., Strand, S., Smith, P., & Fernandes, C. (2007). Intelligence and educational achievement. *Intelligence*, *35*, 13−21.

Dolan, C. V., Colom, R., Abad, F. J., Wicherts, J. M., Hessen, D. J., & van der Sluis, S. (2006). Multi-group covariance and mean structure modeling of the relationship between WAIS-III common factors and sex and educational attainment in Spain. *Intelligence*, *34*, 193−210.

Giedd, J. N., Castellanos, F. X., Rajapakse, J. C., Vaituzis, A. C., & Rapoport, J. L. (1997). Sexual dimorphism of the developing human brain. *Progress in Neuropsychopharmacology and Biological Psychiatry*, *21*, 1185−1201.

Gilmore, J. H., Lin, W., Prastawa, M. W., Looney, C. B., Vesta, Y. S., Knickmeyer, R. C., et al. (2007). Regional gray matter growth, sexual dimorphism, and cerebral asymmetry in the neonatal brain. *The Journal of Neuroscience*, *27*, 1255−1260.

Gur, R. C., Turetsky, B. I., Matsui, M., Yan, M., Bilker, W., Hughett, P., et al. (1999). Sex differences in brain gray and white matter in healthy young adults: Correlations with cognitive performance. *The Journal of Neuroscience*, *15*, 4065−4072.

Gustafsson, J. E., & Balke, G. (1993). General and specific abilities as predictors of school achievement. *Multivariate Behavioral Research*, *28*, 407−434.

Haier, R. J., Jung, R. E., Yeo, R. A., Head, K., & Alkire, M. T. (2005). The neuroanatomy of general intelligence: Sex matters. *Neuroimage*, *25*, 320−327.

Härnqvist, K. (1997). Gender and grade differences in latent ability variables. *Scandinavian Journal of Psychology*, *38*, 55−62.

Hedges, L. V., & Nowell, A. (1995). Sex differences in mental test scores, variability, and numbers of high-scoring individuals. *Science*, *269*, 41−45.

Hu, L., & Bentler, P. M. (1998). Fit indices in covariance structure modeling: Sensitivity to underparameterized model misspecification. *Psychological Methods*, *3*, 424−453.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, *6*, 1−55.

Hulick, P. A. (1998). A structural factor analysis of gender and age differences in cognitive ability. In J. J. McArdle & R. W. Woodcock (Eds.), *Human cognitive abilities in theory and practice* (pp. 247−261). Mahwah, NJ: Erlbaum.

Hyde, J. S., & Linn, M. C. (1988). Gender differences in verbal ability: A meta-analysis. *Psychological Bulletin*, *104*, 53−69.

Irwing, P., & Lynn, R. (2005). Sex differences in means and variability on the progressive matrices in university students: A meta-analysis. *British Journal of Psychology*, *96*, 505−524.

Jackson, D. N., & Rushton, J. P. (2006). Males have greater g: Sex differences in general mental ability from 100,000 17- to 18-year olds on the scholastic assessment test. *Intelligence*, *34*, 479−486.

Jensen, A. R. (1992). Commentary: Vehicles of g. *Psychological Science*, *3*, 275−278.

Jensen, A. R. (1998). *The g factor: The science of mental ability.* Westport, CT: Praeger.

Johnson, W., & Bouchard, T. J. (2007). Sex differences in mental abilities: g masks the dimensions on which they lie. *Intelligence*, *35*, 23−39.

Johnson, W., & Bouchard, T. J. (2007). Sex differences in mental ability: A proposed means to link them to brain structure and function. *Intelligence*, *35*, 197−209.

Johnson, W., Jung, R. E., Colom, R., & Haier, R. J. (2007). Cognitive abilities independent of IQ correlate with regional brain structure. *Intelligence*. doi:10.1016/j.intell.2007.1001.1005

Kano, Y. (2001). Structural equation modeling for experimental data. In R. Cudeck, S. d. Toit, & D. Sörbom (Eds.), *Structural equation modeling: Present and future. Festschrift in honor of Karl Jöreskog* (pp. 381−402). Lincolnwood, IL: Scientific Software International.

Kaufman, A. S., & Horn, J. L. (1996). Age changes on tests of fluid and crystallized ability for women and men on the Kaufman Adolescent and Adult Intelligence Test (KAIT) at ages 17–94 years. *Clinical Neuropsychology*, *11*, 97−121.

Keith, T. Z. (2005). Using confirmatory factor analysis to aid in understanding the constructs measured by intelligence tests. In D. P. Flanagan & P. L. Harrison (Eds.), *Contemporary intellectual assessment: Theories, tests, and issues* (pp. 581–614) (2nd ed.). New York, NY: Guilford Press.

Keith, T. Z., Fine, J. G., Reynolds, M. R., Taub, G. E., & Kranzler, J. H. (2006). Higher-order, multi-sample, confirmatory factor analysis of the Wechsler Intelligence Scale for Children—Fourth edition: What does it measure? *School Psychology Review*, *35*, 108−127.

Keith, T. Z., Kranzler, J. H., & Flanagan, D. P. (2001). What does the Cognitive Assessment System (CAS) measure? Joint confirmatory factor analysis of the CAS and the Woodcock–Johnson Tests of Cognitive Ability (3rd edition). *School Psychology Review*, *30*, 89−119.

Linn, M. C., & Peterson, A. C. (1985). Emergence and characterization of sex differences in spatial ability: A meta-analysis. *Child Development*, *56*, 1479−1498.

Lynn, R. (1999). Sex differences in intelligence and brain size: A developmental theory. *Intelligence*, *27*, 1−12.

Lynn, R., Allik, J., & Irwing, P. (2004). Sex differences on three factors identified in Raven's Standard Progressive Matrices. *Intelligence*, *32*, 411−424.

Lynn, R., Allik, J., & Must, O. (2000). Sex differences in brain size, stature, and intelligence in children and adolescents: Some evidence from Estonia. *Personality and Individual Differences*, *29*, 555−560.

Lynn, R., Fergusson, D. M., & Horwood, L. J. (2005). Sex differences on the WISC-R in New Zealand. *Personality and Individual Differences*, *39*, 103−114.

Lynn, R., & Irwing, P. (2004). Sex differences on the progressive matrices: A meta-analysis. *Intelligence*, *32*, 481−498.

Lynn, R., Irwing, P., & Cammock, T. (2001). Sex differences in general knowledge. *Intelligence*, *30*, 27−39.

Lynn, R., Raine, A., Venables, P. H., Mednick, S. A., & Irwing, P. (2005). Sex differences on the WISC-R in Mauritius. *Intelligence*, *33*, 527−533.

Mackintosh, N. J. (1996). Sex differences and IQ. *Journal of Biosocial Science*, *28*, 559−572.

Maitland, S. B., Intrieri, R. C., Schaie, K. W., & Willis, S. L. (2000). Gender differences and changes in cognitive abilities across the adult life span. *Aging, Neuropsychology, and Cognition*, *7*, 32−53.

McDaniel, M. A. (2005). Big-brained people are smarter: A meta-analysis of the relationship between in vivo brain volume and intelligence. *Intelligence*, *33*, 337−346.

McGrew, K. S. (2005). The Cattell–Horn–Carroll theory of cognitive abilities: Past, present, and future. In D. P. Flanagan & P. L. Harrison (Eds.), *Contemporary intellectual assessment: Theories, tests, and issues* (pp. 136–181) (2nd ed.). New York: Guilford.

McGrew, K. S., & Woodcock, R. W. (2001). *Technical manual. Woodcock–Johnson III.* Itasca, IL: Riverside.

Muthén, B. O. (1989). Latent variable modeling in heterogeneous populations. *Psychometrika*, *54*, 557−585.

Muthén, B. O., Kaplan, D., & Hollis, M. (1987). On structural equation modeling with data that are not missing completely at random. *Psychometrika*, *52*, 431−462.

Nyborg, H. (2003). Sex differences in *g*. In H. Nyborg (Ed.), *The scientific study of general intelligence: Tribute to Arthur R. Jensen* (pp. 5−21). Boston: Pergamon.

Nyborg, H. (2005). Sex-related differences in general intelligence g, brain size, and social status. *Personality and Individual Differences*, *39*, 497−509.

Phelps, L., McGrew, K. S., Knopik, S. N., & Ford, L. (2005). The general (*g*), broad, and narrow CHC stratum characteristics of the

WJ III and WISC-III tests: A confirmatory cross-battery investigation. *School Psychology Quarterly*, *20*, 66−88.

Reiss, A. L., Abrams, M. T., Singer, H. S., Ross, J. L., & Denckla, M. B. (1996). Brain development, gender, and IQ in children: A volumetric imaging study. *Brain*, *119*, 1763−1774.

Reynolds, M. R., Keith, T. Z., Ridley, K. P., & Patel, P. G. (in press). Sex differences in latent general and broad cognitive abilities for children and youth: Evidence from higher-order MG-MACS and MIMIC models. *Intelligence*. doi:10.1016/j.intell.2007.06.003

Rosén, M. (1995). Gender differences in structure, means and variances of hierarchically ordered ability dimensions. *Learning and Instruction*, *5*, 37−62.

Rushton, J. P. (1992). Cranial capacity related to sex, rank, and race in a stratified random sample of 6,325 U.S. military personnel. *Intelligence*, *16*, 401−413.

Sandoval, J. (2003). Review of the Woodcock–Johnson III. In B. S. Plake, J. C. Impara, & R. A. Spies (Eds.), *The fifteenth mental measurements yearbook* (pp. 1024−1028). Lincoln, NE: Buros Institute of Mental Measurements.

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, *7*, 147−177.

Schmithorst, V. J., & Holland, S. K. (2006). Functional MRI evidence for disparate developmental processes underlying intelligence in boys and girls. *Neuroimage*, *31*, 1366−1379.

Schrank, F. A. (2005). Woodcock–Johnson III Tests of Cognitive Abilities. In D. P. Flanagan & P. L. Harrison (Eds.), *Contemporary intellectual assessment: Theories, tests, and issues* (pp. 371–401) (2nd ed.). New York: Guilford.

Steiger, J. H. (1990). Structural model evaluation and modification: An interval estimation approach. *Multivariate Behavioral Research*, *25*, 173−180.

Szabó, C. A., Lancaster, J. L., Xiong, J., Cook, C., & Fox, P. (2003). MR imaging volumetry of subcortical structures and cerebellar hemispheres in normal persons. *American Journal of Neuroradiology*, *24*, 644−647.

Taub, G. E., & McGrew, K. S. (2004). A confirmatory factor analysis of Cattell–Horn–Carroll theory and cross-age invariance of the Woodcock–Johnson Tests of Cognitive Abilities III. *School Psychology Quarterly*, *19*, 72−87.

van der Sluis, S., Derom, C., Thiery, E., Bartels, M., Polderman, T. J. C., Verhulst, F. C., et al. (2007). Sex differences on the WISC-R in Belgium and The Netherlands. *Intelligence*. doi:10.1016/j.intell.2007.1001.1003

van der Sluis, S., Posthuma, D., Dolan, C. V., de Geus, E. J. C., Colom, R., & Boomsma, D. I. (2006). Sex differences on the Dutch WAIS-III. *Intelligence*, *34*, 273−289.

Vernon, P. A., Wickett, J. A., Bazana, G., & Stelmack, R. M. (2000). neuropsychology and psychophysiology of human intelligence. In R. J. Sternberg (Ed.), *Handbook of intelligence* (pp. 245−264). Cambridge: Cambridge University Press.

Voyer, D., Voyer, S., & Bryden, M. P. (1995). Magnitude of sex differences in spatial abilities: A meta-analysis and consideration of critical variables. *Psychological Bulletin*, *117*, 250−270.

Woodcock, R. W., McGrew, K. S., & Mather, N. (2001). *Woodcock–Johnson III Tests of Cognitive Abilities.* Itasca, IL: Riverside.

Yung, Y. F., Thissen, D., & McLeod, L. D. (1999). On the relationship between the higher-order factor model and the hierarchical factor model. *Psychometrika*, *64*, 113−128.